

Bhavan Chithambaram

bhavanchithambaram@gmail.com | bhavanc.github.io/portfolio | linkedin.com/in/bhavan-chithambaram | github.com/BhavanC

PROFESSIONAL SUMMARY

Automotive Control Systems Engineer with over 2 years of industrial R&D experience in electric vehicle development. Specialized in Model-Based Design (MATLAB/Simulink/Stateflow), Battery Management Systems (BMS), implementation of ISO 26262, and advanced control algorithms using Reinforcement Learning. Currently pursuing an M.S. in Commercial Vehicle Technology at RPTU Kaiserslautern.

TECHNICAL SKILLS

Programming: Python, C++, M-script

Frameworks & Tools: MATLAB & Simulink, Stateflow, Git, Virtual validation

Research Areas: Hybrid drives, Battery / Cell testing and validation, Thermal energy management, Image classification, Semantic Segmentation, Sensor Fusion, Reinforcement Learning, ISO 26262

EXPERIENCE

Product Development Engineer

Aug. 2024 – Jul. 2025

Raptee.HV

Chennai, India

- Developed a Digital Twin of a complete battery pack in MATLAB/Simulink to validate cell performance and thermal behavior.
- Developed an algorithm for BMS Passive balancing, reducing heat dissipation by 33%.
- Executed variable charging strategies to enhance cycle life and reduce charging time, achieving 25% faster charging with only 4% loss in energy (Wh).
- Designed and implemented a Vehicle Control Unit (VCU) operating sequence state machine using Stateflow for virtual validation and software-in-the-loop (SIL) testing.

GET-Vehicle Engineer

Jul. 2023 – Aug. 2024

Raptee.HV

Chennai, India

- Characterized electrochemical performance of lithium-ion cells through extensive charge/discharge cycling (C-rate testing), validating Ah and Wh capacity ratings for battery pack design.
- Developed high-accuracy SOC-OCV lookup tables and mapped Internal Resistance (IR) profiles to analyze dynamic voltage variations and thermal efficiency.
- Executed MATLAB/Simulink simulations to identify open-circuit voltage (OCV) hysteresis losses and transient voltage drops across different State of Charge (SOC) operating ranges.
- Designed and implemented control algorithms for Vehicle Control Unit (VCU) throttle mapping and real-time range estimation.

PROJECTS

Sensor Fusion in ACC | EAI 2024

- Optimized object detection reliability in a virtual environment by filtering low-confidence bounding boxes ($< 90\%$), significantly minimizing false positives in tracking data.
- Developed a robust computer vision pipeline for lane boundary detection using Inverse Perspective Mapping (IPM) and histogram-based sliding windows.
- Fused spatial lane tracking data with object detection vectors to accurately identify the Most Important Object (MIO).

FOC in PMSM using RL | IEEE 2023

- Improved reference speed convergence by 10–25 ms in motor control loops by designing and deploying a Twin Delayed DDPG (TD3) reinforcement learning agent.
- Implemented an Artificial Neural Network (ANN) architecture for high-accuracy, sensorless rotor position estimation across dynamic operating conditions.

PUBLICATIONS

- Modelling, Simulation and Virtual Validation of Adaptive Cruise Control (ACC) Algorithm based on Sensor Fusion, EAI, 2024
- Field Oriented Control and speed estimation in PMS motor using RL and ANN, IEEE, 2023

HONORS & AWARDS

2024 – Research Visibility Milestone: Reached 1,900+ technical downloads for the EAI publication on Sensor Fusion in Adaptive Cruise Control.

2023 – Best Paper Award: Selected out of 100+ peer-reviewed submissions at the IEEE conference for excellence in Methodology and Implementation.

EDUCATION

Rheinland-Pflzische Technische Universitt Kaiserslautern-Landau (RPTU) <i>M.S. in Commercial Vehicle Technology</i>	Kaiserslautern, Germany <i>Oct. 2025 – Present</i>
Sri Sairam Engineering College <i>B.E. in Mechanical Engineering</i>	Chennai, India <i>Aug. 2019 – Apr. 2023</i>

LANGUAGES

English: C1 (Professional Working Proficiency)

German: B1 (Currently pursuing)